

Problem 85. Suppose that positive numbers are written in the squares of a 10×10 table. Frogs sit in five squares and cover the numbers in these squares. Alex calculated the sum of all visible numbers and got 10. Then each frog jumped to an adjacent square and Alex's sum changed to 10^2 . Then the frogs jumped again to adjacent squares, and the sum changed to 10^3 and so on. Every new sum became 10 times greater than the previous one. What is the maximum sum Alex can obtain?

Solution. The answer is 10^6 .

First we prove that the sum cannot be greater than 10^6 , that is, 10^7 is unattainable.

In the increase from 10^1 to 10^2 there is an increase of 90. As 5 squares were revealed and 5 taken away, at least one of the revealed squares must have a value strictly greater than $\frac{90}{6} = 15$. However as the total value after is only 100, the square must have value of at most 100. Therefore the square has value in the range $(15, 100)$. Similarly for the other jumps up to 10^6 to 10^7 we must have squares in the ranges:

$$(1.5 \times 10^2, 10 \times 10^2), (1.5 \times 10^3, 10 \times 10^3), \dots, (1.5 \times 10^6, 10 \times 10^6).$$

These 6 squares are in non-intersecting ranges, so they must all be unique. However as there are only 5 frogs, at least one of the squares must be visible initially, which is impossible as even the smallest possible value, 15, creates too big of a sum for the first position. Therefore, Alex cannot get a sum of 10^7 ,

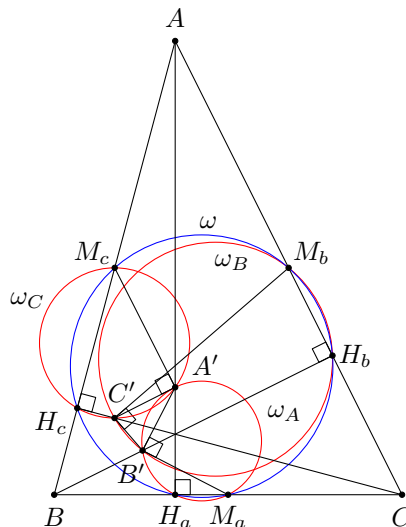
Next we show that 10^6 is attainable. Let $r = \frac{10}{95}$ and let the frogs sit on the first five squares in the upper leftmost corner of the first row and each time let them jump at the same time to the square to the right. Suppose that the frogs cover the numbers f_1, f_2, f_3, f_4, f_5 , in this order in that row, where

$$f_1 = 10^2 - 94r, f_2 = 10^3 - 10^2 + r, f_3 = 10^4 - 10^3 + r, f_4 = 10^5 - 10^4 + r, f_5 = 10^6 - 10^5 + r,$$

and suppose that all the other numbers written in the remaining squares is r . Then these positive numbers written on the squares show that 10^6 is attainable. $\square$

Problem 86. Altitudes AH_a, BH_b, CH_c of an acute triangle ABC meet at H . Let M_a, M_b, M_c be the midpoints of BC, CA, AB , respectively. Points A', B', C' on the segments AH, BH, CH , respectively, are such that $\angle M_a B' A' = \angle M_b C' B' = \angle M_c A' C' = 90^\circ$.

Prove that the lines AC', BA', CB' are concurrent.



Solution. From $\angle M_a B' A' = \angle M_b C' B' = \angle M_c A' C' = 90^\circ$, we see that $M_a H_a A' B', M_b H_b B' C', M_c H_c C' A'$ are all concyclic. Let $\omega_A, \omega_B, \omega_C$ denote three circles that circumscribed those three quadrilaterals, respectively. Now observe that M_a, H_a, M_b, H_b ,

M_c, H_c all lie on the nine-point circle of $\triangle ABC$. Denote this circle by ω . Clearly the radical axes of $\omega_A, \omega_B, \omega$ intersect at C and so C is their radical centre. Consequently, CB' is the radical axis of ω_A, ω_B . Similarly, AC' is the radical axis of ω_B, ω_C and BA' is the radical axis of ω_C, ω_A . Therefore AC', BA', CB' are concurrent at the radical centre of $\omega_A, \omega_B, \omega_C$, and this proves the problem. $\square$

Problem 87. A connected graph is given. Prove that its vertices can be coloured blue and green and some of its edges marked so that

- (i) every two vertices are connected by a path of marked edges,
- (ii) every marked edge connects two vertices of different colour, and
- (iii) no two green vertices are connected by an edge of the original graph.

Solution. Let the connected graph G have n vertices where $n \geq 2$ is a positive integer. We prove the problem by induction on n . For $n = 2$, the result is trivial. Assume now that the problem holds for connected graphs with $n - 1$ vertices. Consider a connected graph G with n vertices and remove a vertex v so that the resulting graph $G \setminus \{v\}$ is still connected (just consider a spanning tree of G and remove a leaf). Then we apply the induction hypothesis to $G \setminus \{v\}$, i.e. its vertices are coloured blue and green and some of its edges are marked satisfying the problem conditions. If there is an edge in G connecting v with some vertex which is now coloured green, we then colour v blue and mark this edge. If all the edges from v connect to blue vertices, we colour v green and mark any one of these edges. In both cases, the marked edges form a connected graph with all the vertices in G , and obviously we have neither new marked edges connecting vertices of the same colour, nor edges of G connecting two green vertices. $\square$

Problem 88. For an integer $n > 1$, let $g(n)$ denote the greatest prime factor of n . A *strange pair* is an unordered pair of distinct primes p and q such that $\{p, q\} = \{g(n), g(n + 1)\}$ for no integer $n > 1$. Prove that there exist infinitely many strange pairs.

Solution. We show that there are infinitely many strange pairs of the form $\{2, q\}$, where q is an odd prime. The Lemma below provides a sufficient condition for such a pair to be strange. For an odd prime q , let $\text{ord}_q(2)$ denote the multiplicative order of 2 modulo q , i.e., the least positive integer s satisfying $2^s \equiv 1 \pmod{q}$.

Lemma. If some primes $2 < q_1 < q_2$ satisfy $\text{ord}_{q_1}(2) = \text{ord}_{q_2}(2)$, then $\{2, q_1\}$ is a strange pair.

Indeed, arguing indirectly, suppose first that $2 = g(n)$ and $q_1 = g(n + 1)$; in particular, $n = 2^k$ for some positive integer k , and $2^k \equiv -1 \pmod{q_1}$. This yields $2^{2k} \equiv 1 \pmod{q_1}$, so $\text{ord}_{q_2}(2) = \text{ord}_{q_1}(2) \mid 2k$. Therefore, $(2^k - 1)(2^k + 1) = (2^{2k} - 1) \equiv 0 \pmod{q_2}$, but $2^k \not\equiv 1 \pmod{q_2}$, hence $2^k \equiv -1 \pmod{q_2}$. So $g(n + 1) \geq q_2$, which is a contradiction.

Similarly, but easier, if $2 = g(n + 1)$ and $q_1 = g(n)$, then $n + 1 = 2^k$, for some positive integer k , so $\text{ord}_{q_2}(2) = \text{ord}_{q_1}(2) \mid k$ and hence $2^k \equiv 1 \pmod{q_2}$. Therefore, $g(n + 1) \geq q_2$, a contradiction and this completes the proof of the Lemma.

Returning to the problem, it remains to show that there exist infinitely many disjoint sets of pairs of primes $q_1 < q_2$ satisfying the conditions in the Lemma.

Let $p = 2r - 1 > 5$ be a prime, and let $N = 2^{2p} + 1$. We prove the following:

- (1) N has at least two distinct prime factors greater than 5; and
- (2) $\text{ord}_q(2) = 4p$ for every prime factor $q > 5$ of N .

Thus, every prime $p > 5$ provides a pair of odd primes satisfying the conditions in the Lemma. Moreover, (2) shows that distinct primes $p > 5$ provide disjoint sets of such pairs, whence the conclusion.

To prove (1), notice that $N \not\equiv 0 \pmod{3}$, and write

$$N = (4 + 1)(4^{p-1} - 4^{p-2} + \dots + 1) \equiv 5p \pmod{25}$$

to infer that $N \not\equiv 0 \pmod{25}$.

Next, write

$$N = (2^p + 1)^2 - 2^{p+1} = (2^p - 2^r + 1)(2^p + 2^r + 1).$$

The two factors are coprime (since they are odd, and their difference is 2^{r+1}), and each is larger than 5. Hence each has a prime factor greater than 5. This establishes (1).

To prove (2), consider a prime factor $q > 5$ of N , and notice that $\text{ord}_q(2) \mid 4p$, since $q \mid N \mid 2^{4p} - 1$. If $\text{ord}_q(2) < 4p$, then either $\text{ord}_q(2) \mid 2p$ or $\text{ord}_q(2) \mid 4$. The former is impossible because $2^{2p} - 1 = N - 2 \equiv -2 \pmod{q}$, the latter because $2^4 - 1 = 15 \not\equiv 0 \pmod{q}$. This establishes (2) and completes the proof. $\square$